

EDITOR : Ramesh Chopra

EDITORIAL : Editorial Secretary
CORRESPONDENCE : Phone: 011-26810601; E-mail: editsec@efy.in
(Technical queries: eflylab@efy.in)

SUBSCRIPTIONS & MISSING ISSUES : Phone: 011-26810601 or 02 or 03
E-mail: support@efy.in

BACK ISSUES, BOOKS, CDs, PCBs etc. : Kits'n'Spares, New Delhi
Phone: 011-26371661, 26371662
E-mail: info@kitsnspares.com

NEWSSTAND DISTRIBUTION : Ph: 011-40596600
E-mail: eflycirc@efy.in

ADVERTISEMENTS NEW DELHI (HEAD OFFICE) : Ph: 011-26810601 or 02 or 03
E-mail: eflyenq@efy.in

MUMBAI : Ph: 022-24950047, 24928520
E-mail: eflymum@efy.in

BENGALURU : Ph: 080-25260394, 25260023
E-mail: eflyblr@efy.in

PUNE : Ph: 08800295610, 09870682995
E-mail: eflypune@efy.in

GUJARAT: : Ph: 079-61344948
E-mail: eflyahd@efy.in

JAPAN : Tandem Inc., Ph: 81-3-3541-4166
E-mail: japan@efy.in

SINGAPORE : Publicitas Singapore Pte Ltd
Ph: +65-6836 2272
E-mail: singapore@efy.in

TAIWAN : J.K. Media, Ph: 886-2-87726780 ext. 10
E-mail: taiwan@efy.in

UNITED STATES : E & Tech Media
Ph: +1 860 536 6677
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at International Print-o-Pack Ltd, C-4 to C-11, Hosiery Complex, Phase-II Extension, NOIDA-201305, Gautam Budh Nagar, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2018. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. Articles that cannot be used are returned to the authors if accompanied by a self-addressed and sufficiently stamped envelope. But no responsibility is taken for any loss or delay in returning the material. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

SUBSCRIPTION RATES

Period	Newstand Price	You Pay	Overseas
Year	(₹)	(₹)	Digital Print
Five	3600	2160	US\$ 20
Three	2160	1515	US\$ 15
One	720	575	US\$ 5 US\$ 100

Please send payments only in favour of EFY Enterprises Pvt Ltd through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com>

Contents



TECH FOCUS

46 Speed up Storage and Access with Solid-State Arrays



A magazine that keeps you up-to-date with technology through easy-to-understand articles

If your product has a mass appeal in the Tech space, then Advertising in eSTYLE would be an ideal opportunity to provide a solution.

To know more, write to nsm@efy.in or call on 011-40596666 (Extn. 214)

SMART WORLD

- 16 Smart Workplace Elements
- 20 Business Opportunities in Hotel Automation Technology
- 24 Towards a Sustainable Future Driven by The Thrust of Organic, Flexible and Printable Solar Power

RESEARCH & DEVELOPMENT

- 50 DISPLAY: Holographic Displays: Making Their Way into Everyday Devices
- 61 SENSORS: Sensors that Add Strength to Aviation and Aerospace
- 64 TEST & MEASUREMENT: Test and Measurement That's Making Aviation Safer
- 68 DESIGN: Development Boards That are Making IoT Designing Simpler
- 74 TELECOM: New Wireless Technologies to Help Meet Aviation Demands
- 91 INTERVIEW: Rishu Sharma, associate research manager, IDC India: "By 2020, over 90 per cent of global enterprises will adopt multi-cloud platforms"
- 92 INTERVIEW: Sumit Joshi, vice chairman and managing director, Philips Lighting India: "IoT-enabled lighting can dramatically reduce energy consumption"

EVERYTHING YOU WANTED TO KNOW ABOUT...

- 54 DEFENCE ELECTRONICS: Optronics Sensors Night Vision Technologies (Part 3 of 6)
- 77 TELECOM: How to Assess Electromagnetic Radiation from Mobilephone Towers
- 84 GREEN TECHNOLOGY: Solar Cold Storage: Green Energy Can Reduce Crop Wastage
- 86 EMBEDDED: Post-Silicon Validation Methodology in SoC (Part 1 of 2)

MAKE IN INDIA

- 26 STARTUP ZONE: IoT-Based Fuel Monitoring System to Bring About Fuel Efficiency for Your Organisation

- 28 STARTUP ZONE: Solar Streetlights with Inbuilt Lithium-Ion Batteries to the Rescue
- 30 STARTUP ZONE: Have an Idea and Want to Become an Entrepreneur, or Become an Entrepreneur and Look for an Idea?
- 38 MARKET SURVEY: Flexible, Printed and Organic Electronics: The Bus India Should Not Miss!
- 43 PROJECT REPORT: Starting an Electronic Counter Making Business

DO-IT-YOURSELF



- 96 Programmable Automatic Bell System
- 100 Wireless Security System using PIR Sensors
- 106 Build Your Own Digital Weighing Machine
- 108 Battery-Powered Night Lamp using an Old LED Bulb
- 110 Two-Channel Wireless Audio Amplifier using Bluetooth and TA8210AH
- 112 Dual Audio Level Shifter and Buffer for ADCs, MCU Kits and Arduino
- 114 Interfacing a Laser LED with Arduino
- 116 Logging Sensor Data in MS EXCEL through MATLAB GUI

NEWS

- 10 Tech News
- 32 Industry News
- 94 New Products

EFY Plus DVD

EFY + DVD: Tools for Engineers, Coders and Media Enthusiasts82
NOOBS 2.8.1: Raspberry Pi Setup Made EasyI
gEDA: A Complete Set of Tools for Electronics DesignIV
KiCad: An Electronics Schematic Design and Layout Creation Suite...VII

Regulars

- 06 Feedback: Your Suggestions
- 08 Q&A: Things You Wanted to Know!